

Fabrication and Integration Process

1. Material Choice Justification :

Why choose Acrylic over MDF? Why PLA instead of Resin? (e.g., weight, durability, aesthetic)

We 3D printed the drone frame out of resin because it was where we had the most control over the density of the product. Since the frame has to be very light, we had to make sure it was crisp and did not have any extra joinery making it heavier.

2. Fabrication Steps :

A documented "Build-log" (e.g., "1. Laser cutting frame," "2. Heat-staking threaded inserts").

- 1) Sketched out the component sizes, especially the motors*
- 2) Downloaded an existing drone frame .stl that matched the motor sizes to save time on measurements*
- 3) Modified the center of the frame to accommodate project specific components or Rhino and Blender*
- 4) Added our initials to the side*
- 5) 3D printed the frame overnight*
- 6) Refined the frame and removed residue and supports*
- 7) Perfectly fit the HCS04 and motors, so we directly mounted those components.*

There was only 1 evolution where the middle was pushed down to create more vertical space in the model.